

The recursive product-density lemma

Claude, with Daniel Murfet

2026-09-07

Abstract

Step 2b of the general- d equal-ratio monomial milestone (Astra #14). With `productIntegral` $\lambda d g$ the recursively defined iterated integral $\int_{(0,1]^d} g(\prod_i t_i) \prod_i t_i^{\lambda-1} dt$, the product-density formula

$$\int_{(0,1]^{m+1}} g\left(\prod_i t_i\right) \prod_i t_i^{\lambda-1} dt = \frac{1}{m!} \int_0^1 g(z) z^{\lambda-1} (-\log z)^m dz$$

holds for every measurable $g: \mathbb{R} \rightarrow [0, \infty]$ (`productIntegral_succ_eq`): the pushforward of $\prod_i t_i^{\lambda-1} dt$ under the product map has density $z^{\lambda-1} (-\log z)^m / m!$ on $(0, 1]$. The induction step scales the inner variable, swaps the order on the triangle and applies the FTC identity of unit 165 (helpers of unit 171). Zero `sorry/axiom`.

1 The things formalised

```
def productIntegral (l : ) : → ( → ENNReal) → ENNReal
| 0, g => g 1
| n + 1, g => ∫ t in Ioc 0 1,
  ENNReal.ofReal (t ^ (1 - 1)) * productIntegral l n (fun z => g (t * z))
def logDensity (l : ) (m : ) (z : ) : := z ^ (1 - 1) * (-Real.log z) ^ m
theorem productIntegral_succ_eq (l : ) (m : ) : g, Measurable g →
  productIntegral l (m + 1) g
  = ENNReal.ofReal (1 / m.factorial)
    * ∫ z in Ioc 0 1, ENNReal.ofReal (logDensity l m z) * g z
```

2 Mathematics

For $m = 0$ the statement is the definition. For the step, the induction hypothesis applied to $z \mapsto g(tz)$ gives an inner integral $\frac{1}{m!} \int_0^1 g(tz) z^{\lambda-1} (-\log z)^m dz$; the substitution $w = tz$ turns it into $\frac{1}{m!} \cdot \frac{1}{t} \int_0^t g(w) (w/t)^{\lambda-1} (\log t - \log w)^m dw$, and $t^{\lambda-1} \cdot \frac{1}{t} \cdot (w/t)^{\lambda-1} = w^{\lambda-1}/t$. Swapping the order on the triangle $0 < w \leq t \leq 1$ and evaluating $\int_w^1 (\log t - \log w)^m / t dt = (-\log w)^{m+1} / (m+1)$ gives the formula with $m!(m+1) = (m+1)!$.

3 Paper-facing meaning

This is the analytic heart of the equal-ratio monomial model in general dimension: it turns the d -dimensional monomial integral into a one-dimensional one with an explicit $(\log)^{d-1}$ density, which is where the log multiplicity d of the Taylor tree for $k = (k_i)$, $(h_i + 1)/(2k_i) = \lambda$ comes from. Remaining: the box bridge from `Fin d` to the recursive integral, the general- λ Gamma/log asymptotic, and the assembly.

4 Lean notes

- Everything in `0ω`: constants leave integrals via `lintegral_const_mul'` (finite constant) or `lintegral_const_mul''` (measurable integrand, needed when the constant is `g w`, possibly infinite).
- `ENNReal.ofReal_mul` must be applied with the exact factorisation in mind; `ring` then handles the commutative rearrangement of the `ofReal` atoms. Pass explicit constants to `lintegral_const_mul'` to control which product is rewritten.
- The joint measurability of the triangle integrand is a one-liner from `Measurable.pow_const`, `Real.measurable_log` and `Measurable.div`.
- After `setLIntegral_congr_fun` the pointwise goals are already beta-reduced (`simp` only makes no progress and errors).